

2(a)(i)	A vector quantity is one that has magnitude and direction.
2(a)(ii)	Velocity Acceleration e.t.c.
2(b)(i)	component W of weight parallel to slope $= 16 \sin 35^\circ = 9.17 \text{ N}$
2(b)(ii)	When the object travels down the slope at constant speed, friction $f = W$ When the object is pulled up the slope at constant speed, $P = f + W = 2W = 18.4 \text{ N}$
2(c)(i)	Impulse $= \Delta p$ $= p_f - p_i$ $= m(v_f - v_i)$ $= (0.22)(3.5 - 2.6)$ $= 0.198 \text{ N s}$
2(c)(ii)	Since the speed of X increases after the collision without a change in direction, Y must be moving in the same direction before the collision as well, in order to exert a force on X acting in the same direction during the collision. By the principle of conservation of linear momentum, $m_x u_x + m_y u_y = m_x v_x + m_y v_y$ $(0.22)(2.6) + (0.40)(3.3) = (0.22)(3.5) + (0.4)v_y$ $v_y = 2.81 \text{ m s}^{-1}$
2(c)(iii)	s = area of trapezium between 1.8 s and 2.0 s $= 0.5(0.2)(3.5 + 2.6)$ $= 0.61 \text{ m}$
3(a)	The fields in both cases are uniform and vertical, such that the gravitational force and